

Plan: GömbSoma-RICCIStim

Adaptive-quadtrees stimulus-driven hypergraph network
with Hodge–Bochner–Ricci geometric propagation

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1 Framing

The earlier GömbSoma-Stim plan ([docs/plans/2026-05-14-gomb-soma-stim/](#)) proposed swapping YOLO’s fixed-grid anchor tiling for a content-driven top- K salience selector. That plan is preserved for design history. **This plan supersedes it** by going further on two axes simultaneously:

1. *Geometric propagation* — replace ad-hoc MLP-salience and flat message passing with a principled differential-geometric stack: Forman-Ricci curvature, Hodge Laplacians at each simplex dimension, and Bochner–Weitzenböck-coupled message passing. This is the mathematical home of "signed-hypergraph vision."
2. *Adaptive multi-scale anchors* — replace uniform-grid patch tiling with **curvature-driven quadtree subdivision**. Patches with high $|\kappa|$ split recursively; uniform-feature regions stay coarse. This gives multi-resolution without YOLO’s FPN-style separate pyramid pathway.

The hypothesis: these two changes together break the rolling-shutter / uniform-grid bottleneck that YOLO-class detectors hit on cluttered small-object scenes.

2 Background: the differential-geometric stack

2.1 Forman-Ricci curvature

For an edge $e = (u, v)$ in a signed hypergraph $H = (V, E, \sigma)$, the Forman-Ricci curvature is

$$\kappa(e) = 2 - \deg(u) - \deg(v) + 2|\Delta(u, v)|$$

where $|\Delta(u, v)|$ is the number of triangles incident on e . Positive κ marks dense / clique-like neighbourhoods; negative κ marks bottleneck edges between otherwise-disjoint regions. Computed once per layer in $O(|E|)$.

2.2 Hodge Laplacian on signed simplicial complex

The signed boundary operator ∂_k maps k -simplices to $(k-1)$ -simplices respecting σ :

$$\partial_k[v_0, v_1, \dots, v_k] = \sum_{i=0}^k (-1)^i \sigma(v_0, \dots, \hat{v}_i, \dots, v_k) [v_0, \dots, \hat{v}_i, \dots, v_k].$$

The Hodge Laplacian at dimension k is

$$\Delta_k = \partial_{k+1}^* \partial_{k+1} + \partial_k \partial_k^*.$$

Δ_0 is the standard graph Laplacian on vertices. Δ_1 acts on edges (edge-flow). Δ_2 acts on triangles (face-flow). The kernel of Δ_k gives the harmonic forms — the topologically invariant features at dimension k .

2.3 Bochner–Weitzenböck identity

Connects the Hodge Laplacian to Ricci curvature:

$$\Delta_1 \omega = \nabla^* \nabla \omega + \text{Ric}(\omega^\sharp)^\flat.$$

On a discrete hypergraph, this means message passing on edges decomposes into a flat-connection diffusion ($\nabla^* \nabla$, ordinary smoothing) plus a curvature correction. **We make this architectural:** each message-passing step has two coupled components, the diffusion term and the Ricci term.

2.4 Curvature-driven graph rewiring (SDRF)

Topping et al. (NeurIPS 2022) showed strongly-negative- κ edges cause exponential over-squashing. Their fix — Stochastic Discrete Ricci Flow (SDRF) — iteratively adds shortcut edges around negative- κ bottlenecks until $\min \kappa$ is bounded away from $-\infty$. We adopt SDRF as a one-time preprocessing step inside the encoder.

3 Adaptive-quadtree anchor selection

3.1 Motivation

YOLO tiles uniform anchors at every position $\times$ scale. Most are wasted on background. Phase 3-V’s PatchGraphBuilder shares this defect: 49 anchors on a 28×28 MNIST image regardless of content. GömbSoma-RICCISTIM replaces this with adaptive subdivision:

3.2 Algorithm

1. Initialise: coarse uniform patches at scale 0 (e.g., 4×4 patches on a 28×28 image $\rightarrow$ 49 level-0 anchors).
2. Compute initial vertex features x_v and Forman κ_v on the current anchor set.
3. Subdivide: every anchor v with $|\kappa_v| > \tau$ (or feature variance $> \sigma_{\text{var}}$, or learned subdivision score $> \theta$) is split into 4 sub-anchors (quadtree).
4. Recompute features and κ on the new anchor set (which is the union of stayed and newly-spawned anchors).
5. Repeat steps 2–4 until max depth D is reached OR total-anchor budget K_{max} is exhausted OR no anchor triggers subdivision.
6. Build the multi-scale signed hypergraph on the final anchor set: *same-scale edges* (spatial adjacency at each depth) + *cross-scale edges* (parent–child quadtree links). Sign every edge by feature-inner-product polarity.

The quadtree IS the hypergraph multi-scale structure. Cross-scale parent-child edges are an extra hyperedge family the Walk / Polygon / Triangle layers consume natively (they are topology-agnostic).

3.3 Budget guarantees

- Max depth $D = 4$ (28×28 MNIST: from 4-pixel patches down to 1-pixel "pixels-as-vertices" extreme).
- Total anchor budget $K_{\max} = 256$ for MNIST, $K_{\max} = 1024$ for Cluttered MNIST (5000 imgs at 5-seed).
- Quadtree depth and total anchor count are deterministic given the curvature threshold τ ; no Optuna.

4 Full architecture

4.1 Pipeline

Stage	Module
1	PatchPatchifier — coarse uniform tiles
2	InitialEmbed — Linear(patch_dim $\rightarrow$ d_{hidden})
3	FormanCurvatureHead — per-vertex κ
4	AdaptiveQuadtree — κ -driven subdivision loop
5	StimulusGraphBuilder — multi-scale signed hypergraph
6	HodgeLaplacian _{0,1,2} — per-dimension propagators
7	SDRF — one-shot curvature-rewiring
8	HypergraphConv-with-Ricci — Bochner-coupled message passing
9	WalkConv / PolygonConv / TriangleConv — existing layers, Ricci-weighted
10	Detection head — per-anchor cls + offset (or per-image cls)

4.2 Bochner-coupled message passing

The base HypergraphConv forward becomes:

$$\text{msg}(c) = \underbrace{\psi_{\pi(c)} \left(\sum_i W_i^{\pi(c)} x_{v_i} \right)}_{\text{flat connection}} + \underbrace{\alpha \cdot (\Delta_k h)_c}_{\text{Hodge term}} + \underbrace{\beta \cdot \sum_i \kappa(e_i) W_i^{\pi(c)} x_{v_i}}_{\text{Ricci correction}}$$

where α, β are learned mixing coefficients. The three terms are the Bochner-Weitzenböck decomposition (flat connection + Hodge Laplacian + Ricci correction). When $\alpha = \beta = 0$ this reduces to the existing Phase 1-3-G layers.

5 Affected files

- `src/hymeko_gomb/soma/vision/forman.py` — FormanCurvatureHead (vertex / edge κ).
- `src/hymeko_gomb/soma/vision/hodge.py` — signed boundary operators ∂_k , Laplacians $\Delta_0, \Delta_1, \Delta_2$.
- `src/hymeko_gomb/soma/vision/quadtree.py` — AdaptiveQuadtree subdivision driver.

- `src/hymeko_gomb/soma/vision/stim_graph.py` — multi-scale signed hypergraph builder.
- `src/hymeko_gomb/soma/vision/sdrf.py` — one-shot curvature-rewiring.
- `src/hymeko_gomb/soma/hg_conv_bochner.py` — optional Bochner-coupled forward as a mixin / wrapper around the Phase-1 HypergraphConv.
- `src/hymeko_gomb/soma/vision/ricci_stim_classifier.py` — MNIST-shaped end-to-end model.
- `src/hymeko_gomb/soma/vision/ricci_stim_detector.py` — Cluttered MNIST per-anchor detector.
- Five matching test files (one per module).
- `experiments/run_gomb_soma_ricci_stim_*.sh`

6 CORE.YAML items touched

None. All modules under `signedkan_wip/`.

7 Phased rollout (one per session, CLAUDE.md §6.5 #11)

1. `FormanCurvatureHead` — κ on the patch graph; unit tests including the analytic check $\sum_e \kappa(e)$ matches Gauss–Bonnet on a closed cycle.
2. `AdaptiveQuadtree` — κ -driven recursive subdivision; unit tests including determinism and depth bound.
3. `HodgeLaplacian0/1/2` — signed boundary ∂_k and Hodge Δ_k ; unit tests including $\partial_k \partial_{k+1} = 0$ and the Hodge decomposition theorem on small examples.
4. `Bochner-coupled HypergraphConv` — α, β mixing of flat / Hodge / Ricci terms; unit tests including: $\alpha = \beta = 0$ reproduces Phase 1-3-G exactly.
5. `StimulusGraphBuilder` — multi-scale signed hypergraph from quadtree + cross-scale edges; unit tests.
6. `SDRF rewiring` — iterative curvature-flow edge rewiring; unit tests.
7. End-to-end MNIST classifier; smoke train.
8. End-to-end Cluttered MNIST detector + falsification battery (target ≥ 0.5 mAP50).

Each phase ships a working module under unit-test coverage. No phase combines.

8 Test strategy

Standard for each phase plus:

- *Gauss–Bonnet check* for κ : on a closed simply-connected region, $\sum_e \kappa(e) = 2\pi\chi$ (χ = Euler characteristic). Pinned at low tolerance on a small graph.

- *Hodge decomposition* on a 5-vertex example: any 1-form ω decomposes uniquely as harmonic + gradient + curl; verified by reconstruction.
- *Boundary chain rule* $\partial_k \partial_{k+1} = 0$ pinned exactly.
- *SDRF monotonicity* after rewiring, $\min_e \kappa(e)$ is strictly higher (bottleneck removed).
- *Phase-1-3-G reproducibility*: with $\alpha = \beta = 0$, the Bochner-coupled HypergraphConv produces bit-identical output to Phase-1’s vanilla forward.

9 Falsification battery

Reference	Score (mAP50)	GömbSoma-RICCIStim prediction
HyMeYOLO +kcycle (broken)	0.204	beat strongly
HyMeYOLO +ricci+kcycle	0.235	beat strongly
HyMeYOLO +ricci-mod (positive)	0.723	match or beat
HyMeYOLO boxes+circles	0.715	match or beat
Phase 3-V rolling-shutter GömbSoma	TBD	beat

Target: ≥ 0.72 mAP50 on Cluttered MNIST at comparable parameter count.

10 Performance budget

- Peak RSS ≤ 14 GB (cgroups 16 GB).
- Per-seed wall on Cluttered MNIST (5000 imgs, 50 epochs): ≤ 30 min on the RTX 2070 SUPER (HyMeYOLO baseline ~ 13 min; Ricci-Stim adds Hodge SpMM and quadtree).
- GPU memory ≤ 7 GB.
- Full falsification battery (5 seeds): ≤ 3 h total.

11 Risk anticipation

- *Quadtree training instability.* The subdivision decision is non-differentiable. Mitigation: use a soft subdivision score (sigmoid on a learned function of κ and feature norm) with a straight-through estimator; hard subdivide at inference.
- *Hodge Laplacian on large simplicial complexes.* Δ_2 on Cluttered MNIST may be expensive. Cap triangle count per anchor; rely on sparsity.
- *Bochner coupling untraced ground.* The three-term forward has been studied in the GNN literature (CurvGN, Topping 2022, sheaf neural networks) but never with explicit signed simplicial structure. Mitigation: ablation in phase 4 ($\alpha = 0$, $\beta = 0$, both, neither) to isolate contributions.
- *HyMeYOLO baseline reproducibility.* Use identical training protocol (data split, augmentation, epochs) to the prior HyMeYOLO runs — this is the basis of the paired comparison.
- *Empty-plan-dir hygiene.* If shelved, plan dir is deleted before commit.

12 Decision tree

- ≥ 0.72 mAP50 $\Rightarrow$ paper-grade YOLO-class result; write up as a contribution: *adaptive-quadtree stimulus-driven hypergraph network with Hodge-Bochner-Ricci geometric propagation*.
- $0.5\text{--}0.72 \Rightarrow$ stimulus-driven hypothesis confirmed but not at YOLO scale; document and continue.
- $0.235\text{--}0.5 \Rightarrow$ partial confirmation; ablate (which of Hodge / Ricci / quadtree / σ -balance contributes).
- $\leq 0.235 \Rightarrow$ stimulus-driven hypothesis falsified; document as negative result.

13 Why this is the right ambition

- *Differential geometry is the right framework for vision.* CNN convolution is a clumsy discretisation of Δ_0 heat flow on a planar grid. Hodge-aware hypergraph networks are the natural generalisation: irregular topology, signed structure, arbitrary simplex dimensions.
- *Adaptive multi-scale without FPN.* Quadtree subdivision gives multi-resolution detection from a single forward pass over one shared hypergraph. No separate pyramid pathway.
- *Cartwright–Harary balance becomes a region-coherence criterion.* The σ -product on triangles tells us *which combinations of regions form a coherent object* — exactly what YOLO can’t express.
- *Empirically anchored.* HyMeYOLO already showed Ricci alone works on Cluttered MNIST (0.723). The hypothesis is that Ricci + Hodge + quadtree + σ -balance can match it at fewer parameters with better generalisation.

14 Bottom line

GömbSoma-RICCI_{STIM} combines three substantive architectural moves with the existing GömbSoma layer stack: (a) adaptive-quadtree anchor selection driven by Forman-Ricci curvature; (b) full Hodge-Bochner- Ricci geometric message passing; (c) SDRF curvature-rewiring against over-squashing. The plan is phased, each phase produces a tested module, and the falsification battery has concrete prior baselines. Target: beat or match HyMeYOLO +ricci-mod (0.723 mAP50) on Cluttered MNIST. A confirmed result is a paper-grade YOLO-class architectural contribution.